

Venkata Satya Sreeya Chandrapati

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SUMMARY

Data Analyst with strong data science fundamentals, combining **automation, dashboarding, and predictive modeling** to deliver measurable outcomes. Built classification models achieving **~95% accuracy and >0.92 ROC-AUC**, and developed **Tableau/Power BI** reporting to convert raw data into decision-ready insights. Known for streamlining manual workflows using **Excel/Google Sheets + Apps Script/Python**, improving reporting speed and consistency.

SKILLS

- **Programming & Query:** Python, SQL, R, SAS
- **Analytics & BI:** Excel, Power BI, Tableau, Google Sheets
- **ML & Data Science:** Scikit-learn, Statistics/EDA, Classification, Model Evaluation (AUC, F1, ROC)
- **Automation & Workflow:** Google Apps Script, Jupyter, Git, GitHub, VS Code

EDUCATION

University of Texas at Arlington

Master of Science (M.S) in Applied Statistics and Data Science (CGPA: 4.0)

Arlington, Texas

Aug 2025 - May 2027

- **Relevant Courses:** Statistical Theory and Application, Statistical & Scientific Computing, Principle of Data Science, Machine Learning with Applications, Applied Time Series Analysis, Advance Regression Analysis.

WORK EXPERIENCE

The University of Texas at Arlington

Teaching Assistant

Hyderabad, Telangana

Feb 2026 - Present

- Support instructional delivery for DATA 3441: Stats for Data Science 1, focusing on fundamental concepts including descriptive statistics, probability theory, and inferential statistics.
- Facilitate weekly laboratory sessions for a cohort of students, guiding hands-on application of statistical methods and data visualization using SAS software.
- Assist in the evaluation of student performance by grading homework, quizzes, and lab assignments for a 4-credit hour curriculum.
- Provide academic support and technical guidance on topics such as least-squares regression, sampling distributions, and hypothesis testing.

Deccan AI

Project Associate

Hyderabad, Telangana

Mar 2025 - July 2025

- Managed multiple assessment-content delivery streams end-to-end (planning > execution > QA), consistently meeting timelines and quality thresholds across domains.
- Built and managed an SME + freelancer delivery pipeline (sourcing, onboarding, task assignment, review), improving throughput and reducing rework through clearer acceptance criteria.
- Designed evaluation rubrics and streamlined review workflows with cross-functional stakeholders, shortening iteration cycles and improving consistency across reviewers.
- Implemented QA checklists and audit routines to standardize quality, increasing content reliability and reducing variance in reviewer outcomes.

AI Annotator

Nov 2024- Mar 2025

- Created and annotated high-quality training datasets for model improvement, following strict labeling guidelines to maximize signal quality.
- Supported RLHF-style refinement by generating structured feedback on model responses, improving response quality and reducing recurring failure patterns.
- Maintained dataset consistency through QA sampling and error logging, improving label accuracy and minimizing noise in training data.

Tata Advanced Systems (TASL)

Data Analytics Intern

Hyderabad, Telangana

Jan 2024 - Apr 2024

- Debugged and resolved Google Apps Script failures in automated workflows, improving process stability and reducing disruptions to recurring operations.
- Built Google Sheets reporting logic (custom formulas + structured templates) to analyze large datasets and generate management-ready summaries for project decisions.
- Developed Tableau dashboards to track operational KPIs and organizational metrics, improving visibility into performance drivers and enabling faster stakeholder reporting.
- Automated invoice delivery using Apps Script, reducing manual follow-ups and improving turnaround consistency for internal processing.

PROJECT EXPERIENCE

University of Texas at Arlington

Correlation of Kinetic and Thermal Energies of Protons in Space Plasma

Arlington, Texas

Jan 2026 - Present

- Developing data analysis workflows under the supervision of Prof. Anton Artemyev (Department of Physics, UTA).
- Investigating the mechanisms of fast plasma flow thermalization within the Earth's magnetosphere and the near-Earth solar wind.

- Analyzing high-resolution plasma measurements from the NASA THEMIS and MMS (Magnetospheric Multiscale) missions to identify energy conversion patterns.
- Quantifying the correlation between kinetic and thermal energy states to better understand energy dissipation in collisionless plasmas.

PAW-ANALITICS: Predictive Operations Framework

Mar 2026 - May 2026

- Addressed operational inefficiencies and overcrowding at animal shelters by leveraging the Dallas Animal Shelter public dataset (34.8K records).
- Developed a dual-model machine learning pipeline to predict animal outcomes and estimate Length of Stay (LOS) to optimize resource allocation.
- Engineered 13 critical features and implemented advanced ensemble models, including a Random Forest classifier and XGBoost regressor, while mitigating data leakage to ensure model integrity.
- Achieved a high classification accuracy of 91.5% ROC-AUC and reduced stay-duration prediction error to a Mean Absolute Error (MAE) of 5.56 days.

Hierarchical Analysis of Structural Borrowing Costs in Texas Municipal Debt

Mar 2026 - May 2026

- Analyzed 17K+ Texas municipal debt issuances using Hierarchical Linear Models (R) to identify borrowing cost drivers, improving model fit (AIC) from -4566 to -4792.
- Engineered Interest-to-Principal Ratio (IPR) metric to quantify a 0.06 premium for Revenue vs. General Obligation bonds, controlling for issue purpose and size.
- Led end-to-end data preprocessing and validation, utilizing GVIF multicollinearity checks and residual diagnostics to ensure robust model inference and reliability.

Capital Bikeshare Demand Forecasting & Supply Optimization

Mar 2026 - May 2026

- Seasonal and intraday demand volatility made it difficult for transit stakeholders to plan maintenance and redistribution.
- Identify optimal operational windows and supply levels for a large-scale urban bike network.
- Conducted Exploratory Data Analysis (EDA) to isolate a bimodal weekday commuter cycle and a unimodal weekend leisure plateau; utilized Prophet to audit the impact of US Federal Holidays and weather regressors on demand.
- Mathematically validated a zero-disruption maintenance window (3-4 AM) and established a data-driven redistribution blueprint that increased system reliability across 700+ daily summaries.

Multiclass Classification of Liver Disease

Nov 2025 - Dec 2025

- Built a multiclass clinical classification pipeline to segment patients into 4 diagnostic categories using demographic + biochemical markers.
- Engineered preprocessing in R (cleaning, encoding, 99th-percentile outlier capping, z-score scaling) to standardize clinical features for modeling.
- Reduced multicollinearity by applying PCA and used ANOVA/effect-size analysis to identify high-signal biomarkers (AST, BIL, GGT).
- Trained and compared MLR/LDA/QDA; best model achieved 94.91% accuracy and 0.79 macro F1 (LDA on PCA features).

Predicting Company Bankruptcy

Nov 2025 - Dec 2025

- Built an early-warning bankruptcy classifier using 6,819 Taiwanese firms with 95 financial indicators and a 3.23% bankruptcy rate.
- Addressed severe class imbalance by benchmarking SMOTE against class weighting, optimizing recall on the minority class without collapsing precision.
- Reduced predictors from 95 to 6 using VIF-based multicollinearity removal + forward selection (BIC), improving stability and interpretability.
- Achieved ROC-AUC > 0.92, maintained ~95% accuracy, and tuned thresholds to capture 57% of bankruptcy cases.

Identifying Key Determinants of Heart Disease

Nov 2025 - Dec 2025

- Built a SAS-based clinical analytics pipeline on the UCI Heart Disease dataset to identify statistically significant drivers of heart disease risk.
- Cleaned and encoded mixed-type variables and performed EDA to validate distributions and modeling readiness prior to inference.
- Ran inferential tests (t-tests, ANOVA, chi-square) to screen predictive signals and reduce noise before modeling.
- Fit a stepwise logistic regression model achieving ROC-AUC = 0.9365 and 87% sensitivity, isolating high-signal predictors (thalassemia, vessel blockage).

MVSR Engineering College

Hyderabad, Telangana

Dehazing Image and Video with Object Detection

Oct 2023 - June 2024

- Built an image & video dehazing pipeline using Dark Channel Prior + structural priors to improve visibility under haze without major detail loss.
- Integrated YOLOv5 object detection to measure downstream detection improvements after dehazing on real-world frames.
- Optimized the pipeline for high-fidelity video processing, reducing artifacts and preserving edge structure for detection readiness.
- Deployed an interactive Streamlit app enabling users to upload media for dehazing and real-time object detection.

Cancer Data Analysis Dashboard with Power BI

Mar 2024 - Apr 2024

- Analyzed global cancer incidence and mortality trends (2000–2019) and translated findings into stakeholder-ready visuals.
- Built an interactive Power BI dashboard with filters by year, age group, and cancer type to enable drill-down analysis.
- Identified multi-year trend shifts and age-specific prevalence patterns to support high-level comparative insights.

- Packaged the dashboard as a reusable reporting artifact for fast exploration and presentation.

Liver Disease Prediction using Machine Learning

Mar 2023 - Aug 2023

- Built and evaluated multiple ML classifiers on the Indian Liver Patient Dataset (ILPD) to predict liver disease risk from clinical features.
- Compared 8 models (LogReg, KNN, SVM, RF, AdaBoost, GBM, DT, NB) using standardized preprocessing and consistent evaluation.
- Selected Logistic Regression as the best-performing model, achieving 75% accuracy on held-out evaluation.
- Deployed a Flask web app to collect clinical inputs and return predicted liver-disease risk for user-facing assessment.

LEADERSHIP EXPERIENCE

IEEE, MVSREC

Hyderabad, Telangana

Women in Engineering Chairperson

July 2023 - Jun 2024

- Planned and delivered technical workshops and networking sessions, iterating event formats using participant feedback to improve execution quality and relevance.
- Designed and ran a structured student–alumni mentorship program, supporting career guidance, project planning, and interview preparation for participating members.
- Executed STEM outreach programs for school students by designing age-appropriate technical sessions, increasing early exposure to engineering concepts with a focus on women in STEM.
- Coordinated industry speaker sessions and career clinics, expanding member exposure to technical roles, hiring expectations, and resume-level preparation.

LICENSES AND CERTIFICATIONS

Large Language Model Operations (LLMOps), Introduction to R Programming for Data Science, Academy Graduate - AWS Academy Machine Learning Foundations, Data Science with Python, AWS Academy Graduate - AWS Academy Data Engineering, Power BI for Beginners, Smart Coder